

## Article

# System Layout and Grasp Efficiency Optimization for a Multirobot Waste Sorting System

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**Abstract:** The transition towards a circular economy, as outlined in the European Union's Green Deal, requires the development of industries dedicated to recycling and material recovery. Within this context, the recycling of plastic and packaging waste is critical in reducing greenhouse gas emissions. Traditional pick-and-place systems encounter significant challenges when applied to heterogeneous waste streams due to the variability in shape, weight, and material properties of the processed materials. To address these challenges, this research proposes a heuristic to optimize the use of multiple gripper systems within a multirobot multigripper sorting setup, with the goal of both maximizing sorting efficiency and recovery rates in PPW recycling. Therefore, the performance of grippers on specific PPW objects, materials and shapes is quantitatively assessed by measuring the grasp efficiency. This grasp efficiency is incorporated into the proposed scheduling heuristic and used to assign the PPW objects to the different available robots, taking into account the position of the object with respect to the robot and the gripper installed on the robot. This heuristic is then evaluated and benchmarked through simulations considering the sorting system design and the waste stream composition based on a real-world portable robotic material recycling facility. The findings demonstrate substantial improvements in picking efficiency of up to 3.6% and pick rates up to 37.5%, underscoring the potential of advanced heuristic algorithms in robotic waste sorting systems. Future work will focus on refining gripper designs and exploring predictive algorithms to further enhance grasp success rates.

**Keywords:** gripper efficiency; operational optimization; pick and place; robotic sorting



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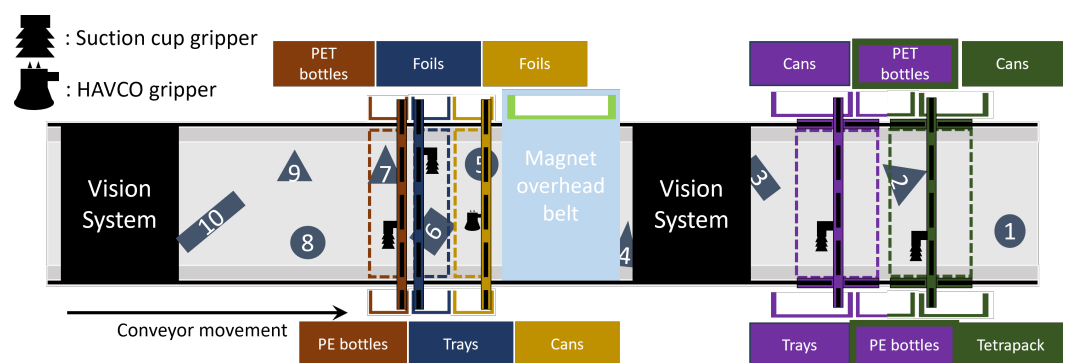
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## 1. Introduction

As the European Union's Green Deal sets new challenges to transition industries toward greater circularity [1], new industries have emerged, focusing on the recycling of materials. Among these, the sorting of materials plays a crucial role in facilitating recycling and reducing reliance on the production of new primary materials. Tallentire et al. [2] evaluated various plastic and packaging waste (PPW) collection and processing scenarios, demonstrating that these processes, which account for 2.8% of the total greenhouse gas emissions in Europe, can be improved. By adapting the collection and processing methods for PPW, these emissions can be reduced by 0.37–0.47%. Part of the collection process is the sorting of PPW. Sorting processes traditionally rely on a combination of sensors and pneumatic separation techniques [3]. However, alongside the growing integration of

robotics in traditional manufacturing industries, robots are increasingly being incorporated in state-of-the-art sorting equipment [4]. It is common for a robotic sorting system to be composed out of one or two robots in addition to the conveyor belt and vision system. Mostly Cartesian, delta, or Selective Compliant Assembly Robot Arm (SCARA) robots are used [5–11]. Delta and SCARA robots are commonly used when relatively low payloads and fast cycle times are required. Cartesian robots are generally more cost effective in terms of construction and can carry more payload at the cost of speed. In manufacturing and logistics industries, pick-and-place (PnP) systems typically handle products with known shapes, weights, and materials. In contrast, PnP systems in PPW sorting must process a non-homogeneous product stream, where each item varies in shape, material composition, and weight. This variability presents challenges and opportunities in designing sorting systems for material recovery facilities (MRFs). Unlike traditional industries where profit is driven by the number of picked products, in PPW sorting, profitability is achieved by maximizing the mass sorted while limiting the loss of material quality due to impurities in the sorted fractions [12]. Additionally, due to the abundance of waste to sort, sorting systems mostly operate in an overloaded condition, especially since in state-of-the-art systems, only one or two robots are integrated to pick a minority fraction from the waste material.

However, by integrating more, lower-cost, robots, a bigger fraction of the material can be targeted [13]. Another key challenge is selecting a universal gripper system capable of successfully grasping this wide range of products with diverse characteristics. Therefore, employing multiple different gripper devices presents an opportunity to enhance efficiency by matching the most appropriate gripper to each PPW object. Aside from the selection of grippers for a robotic manipulation task in manufacturing and logistic industries [14], prior work proposed a framework to select (a suitable set of) grippers by linking gripper technologies to the constantly changing waste stream [15]. Prior work shows the usage of multiple grippers on one robot where, based on computer vision, regions of interest for grasping with the available grippers are detected and used to decide which gripper is most suitable to pick the specific available objects [16]. Instead, the different gripper technologies can be installed on multiple different robots, allowing the use of multiple different gripper systems at the same time. To maximize grasp success by effectively utilizing multiple gripper systems, this study introduces a novel optimization algorithm, building on prior work, allowing the use of multiple different grippers in a more efficient way within the proposed system (Figure 1).



**Figure 1.** Schematic overview of the pick-and-place scheduling problem with time windows. Each picking robot has its own designated workspace represented by a colored box on the conveyor belt. After detection of waste objects with different material properties and shapes, the objects are classified and labeled with a unique ID (1–10), allowing the scheduling algorithm to optimize.

The proposed heuristic maximizes the number of successful picks while operating the PPW MRF by assigning the best available gripper and robot candidate to every waste object that needs to be sorted.

The objectives of this study include implementing, validating, and comparing the proposed heuristic against commonly used task scheduling approaches in PnP systems, with a focus on optimizing grasp efficiency and the number of picks within a given time frame, expressed as picks per minute (PPM). Additionally, this study evaluates the impact of integrating the proposed heuristic on a real-world industrial use case.

The structure of this article is as follows: Section 2 provides an overview of the materials and methods used in this study, including the simulation environment, use case, process time estimation, gripper efficiency measurement, and heuristics. The performance of the grippers is examined in Section 2.1, while the estimation of the time required to pick each object is discussed in Section 2.2. The assignment of objects to specific robots can be optimized using intelligent scheduling algorithms, which are explored in Section 2.3. Section 3 details the experiments conducted, while the results and insights are discussed in Section 4. Finally, some conclusions and suggestions for future work are presented in Section 5.

## 2. Materials and Methods

A schematic representation of the proposed system with multiple robots and different gripper systems is shown in Figure 1. In this configuration, objects that need to be picked and sorted enter the machine over the conveyor belt and are subsequently detected by the vision system. The conveyor belt then transports the objects to the robots' designated workspaces. The robots are generally positioned alongside or above the conveyor belt, with each robot assigned to a specific, non-overlapping workspace. As shown in Figure 1, a second vision system is installed in the system. However, this is only used to relocate objects after possible relocation by the first three robots. Due to the fact that the objects do not move a considerable amount, the impact of the possible relocation is negligible. The proposed system comprises five robotic sorting units, each developed to be low-cost, represented by a different color in Figure 1. The first three robotic sorting units (brown, blue, and yellow) are designed with a linear, servo-actuated rail for horizontal motion and a pneumatic-actuated linear actuator for vertical motion. This setup is referred to as a 1.5-degree-of-freedom (DOF) robot, as the vertical motion lacks precise position control. The fourth robot (purple) features servo-actuated linear rails in three directions: horizontal perpendicular to the conveyor belt, horizontal parallel to the conveyor belt, and vertical. This configuration is further referred to as a 3-DOF robot. Similarly, the fifth robot (dark green) is termed a 2.5-DOF robot due to its two horizontal servo-actuated linear motions and a pneumatically actuated vertical motion. After the third robot, a magnetic overhead conveyor belt is positioned in the middle of the system to remove ferrous materials. Once the ferrous materials are removed, a second vision system is installed to redetect the remaining objects before they are conveyed to the last two robots. This step ensures that the objects are accurately tracked, even if they were displaced during processing off surrounding objects by the first three robots. However, it is important to note that objects not picked by the first three robots, due to a failed pick attempt, will not be reattempted for picking after redetection. To make the system portable, the system is developed to fit in a shipping container, making it a portable robotic MRF (pr-MRF). In contrast with the state of the art in robotic sorting installations, this pr-MRF can be deployed for a limited time in remote locations where the realization of a permanent MRF is not possible or close to the source of PPW, for instance, a big event, to limit the transportation of waste to MRFs. The PPW considered in this study originates from the packaging waste bins collected by

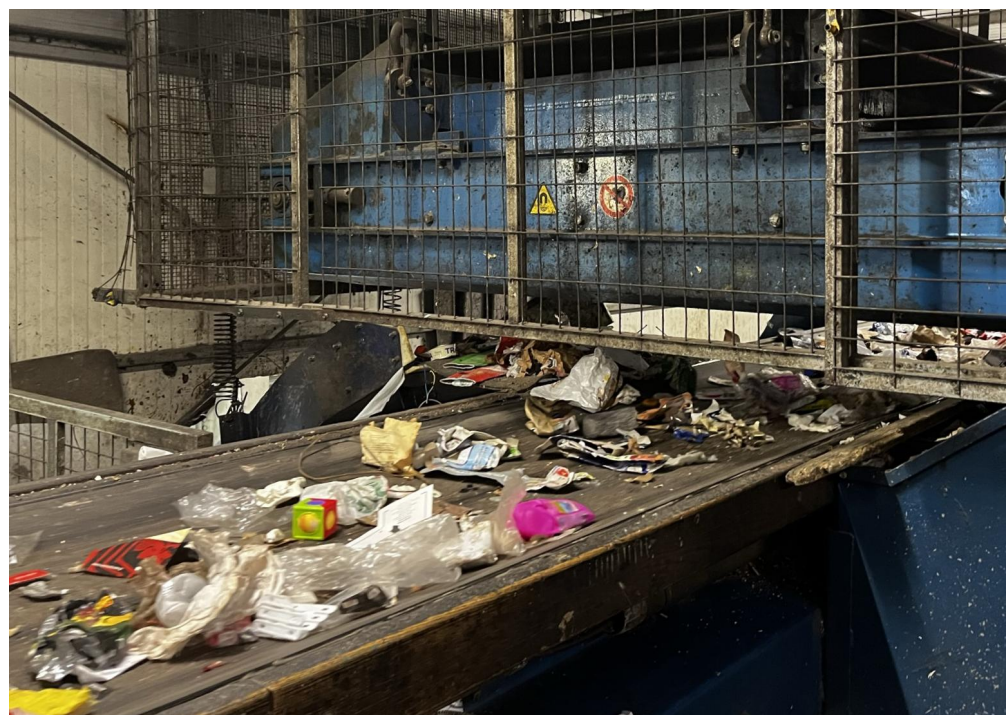
a communal system in Heraklion, Crete. The composition of the waste stream handled in this case is shown in Table 1. It can be observed that no average mass is available for multilayer films, as it has not been reported for this class. The waste stream includes objects with varying degrees of pollution and deformation, resulting in a highly heterogeneous waste stream.

**Table 1.** Waste stream composition—materials.

| Material        | RECLAIM Target Stream (t) | RECLAIM Target Stream (%) | Average Recyclable Weight (g/Item) |
|-----------------|---------------------------|---------------------------|------------------------------------|
| TETRAPACK       | 74                        | 4.2%                      | 39                                 |
| PET             | 597                       | 33.5%                     | 33                                 |
| PE—bottles      | 82                        | 4.6%                      | 104                                |
| PE—film         | 431                       | 24.2%                     | 48                                 |
| Multilayer—film | 23                        | 1.3%                      |                                    |
| PP              | 211                       | 11.9%                     | 40                                 |
| PS              | 30                        | 1.7%                      | 40                                 |
| AL              | 96                        | 5.4%                      | 15                                 |
| FE              | 236                       | 13.3%                     | 87                                 |

An example of the pollution rate and heterogeneity is shown in Figure 2. It is observable how the physical appearance of the objects is changed due to pollution and deformation, how different materials are interlocked due to the deformation, and how collections of material are present that obstruct the visibility for computer vision algorithms and grasping with robots.

Given that a variety of objects will pass through the workspaces of multiple robots, different robots with potentially different grippers can be assigned to pick specific objects. Since objects move through the robots' workspaces linearly and at a constant velocity, and because the workspaces have limited sizes, each object is available to a specific robot for only a brief period. This scenario presents a time-dependent pick-and-place scheduling problem with time windows (TD-PnP-TW). The proposed heuristic will be validated through a simulation study based on the industrial use case.



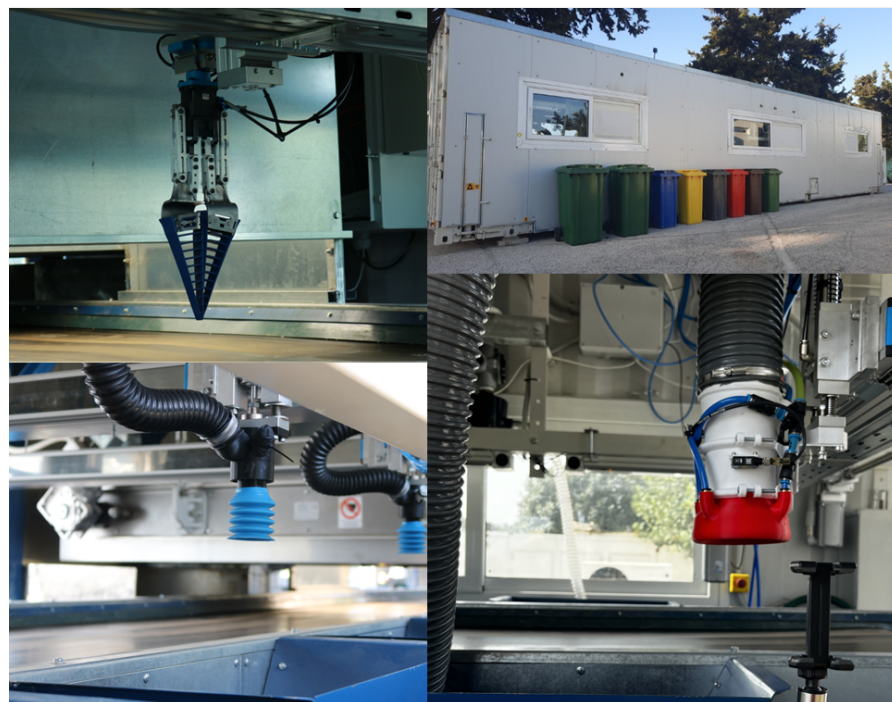
**Figure 2.** Example of waste material which is treated in pr-MRF in Heraklion, Greece. Different types of waste with a high level of pollution and variety in shape and material are visible.



### 2.1. Gripper Efficiency Estimation

In prior work, the authors performed a literature review on the different available grasping techniques with the same use case and sorting system layout in mind. The literature review showed that impactive, ingressive, and astrictive conventional grippers and actuation-controlled soft grippers with both passive structures and inflatable structures could be interesting for the manipulation of waste [15]. These picking principles in the application of waste sorting could result in 3 hardware configurations: finger grippers, grippers that astrict with suction, and grippers that use intrusive elements to fixate an object. An example of intrusive elements is pins that permeate the surface of an object. However, since intrusive elements are rather fragile and uncommon in the state of the art due to the polluted nature of the PPW materials, this grasping principle is not further considered.

The other two options are more common in the state of practice of waste sorting systems. Finger grippers are used in robotic sorting systems for the sorting of construction waste that is bigger and heavier than PPW [7,10]. However, some PPW objects can be big, for instance, detergent bottles. Therefore, a finger gripper equipped with passive-actuation-controlled soft structures [17] is developed and installed for evaluation. Another common gripper principle which defines the norm in the waste sorting industry is astrictive grasping with suction cup grippers [5,6,8,9,11]. Additionally, a novel astrictive gripper type, the HAVCO gripper, which has been proven effective on aluminum waste objects [18], is also tested on the PPW waste stream in the pr-MRF. The HAVCO grasping principle leverages the Coanda effect to produce a large secondary airflow using a relatively small primary airflow generated by pressurized air. This secondary airflow serves a dual purpose: initially, it is used to astrict the object and draw it into the gripper. Once the object is within the gripper, the secondary purpose is to transport the object through a tube, ultimately delivering it to the designated sorting bin. Transporting the objects through a tube supersedes the need for robotic motion towards the sorting bin, significantly reducing the overall picking time and enhancing system efficiency. Figure 3 shows all the gripper types and the pr-MRF setup considered in these experiments.



**Figure 3.** The pr-MRF and the installed grippers are shown: the parallel gripper (**top-left**), the suction cup gripper (**bottom-left**), the pr-MRF (**top-right**), and the HAVCO gripper prototype (**bottom-right**).

To allow the scheduling algorithm to assign the gripper with the highest success rate to each of the PPW objects, the statistical performance of each gripper on all of the object types has to be defined. To achieve this, the authors of this study categorized the PPW stream into subclasses of waste objects. Each gripper was then selected and tested on these subclasses within the pr-MRF in Heraklion.

In the pr-MRF setup, the first two 1.5-DOF robots (brown and blue in Figure 1) are equipped with suction cup grippers. To grasp with the suction cups, a heuristic is used to position the grippers on the surface of the PPW objects above the centroid of the object. The third 1.5-DOF robot (yellow in Figure 1) is equipped with a HAVCO gripper. In contrast to the suction cup grippers, the HAVCO gripper does not have to be positioned on the object's surface. The gripper only has to be positioned within one cm above the object's surface above the centroid of the object. Because this gripper uses high airflow to remove objects from the conveyor belt, the HAVCO gripper is installed last among the first series of robots, just before the second detection unit. While the high airflow is effective for picking specific waste objects, it could potentially unintentionally displace nearby objects. However, the presence of the second detection system mitigates this issue, ensuring that any accidental movement of objects does not impact the overall sorting process. The 3-DOF robot (purple in Figure 1) is equipped with a finger gripper; since finger-gripper grasping is relatively slow and therefore requires the robot to follow the object for a brief moment, this is called tracking. During the tracking motion, the gripper can be closed and the pressure in the pneumatic actuator can build to allow for sufficient grasping force by the gripper. In addition to aligning the gripper to the centroid of the object, the two-finger gripper needs to be aligned with the principal component of the PPW object. This cannot be realized with the installed 3-DOF Cartesian robot. For this reason, an additional joint needs to be installed. To minimize cost, complexity of control, and required payload, a pneumatic swivel actuator is installed. This, however, compromises the functionality of the gripper. Due to the nature of the pneumatic swivel actuator, the gripper can only be rotated to either the 0-degree position or the 90-degree position. This introduces a suboptimal working condition for the gripper. The final 2.5-DOF robot (dark green in Figure 1) is equipped with a suction cup gripper. This strategic placement and gripper selection aims to optimize the picking efficiency and accuracy across the diverse PPW stream.

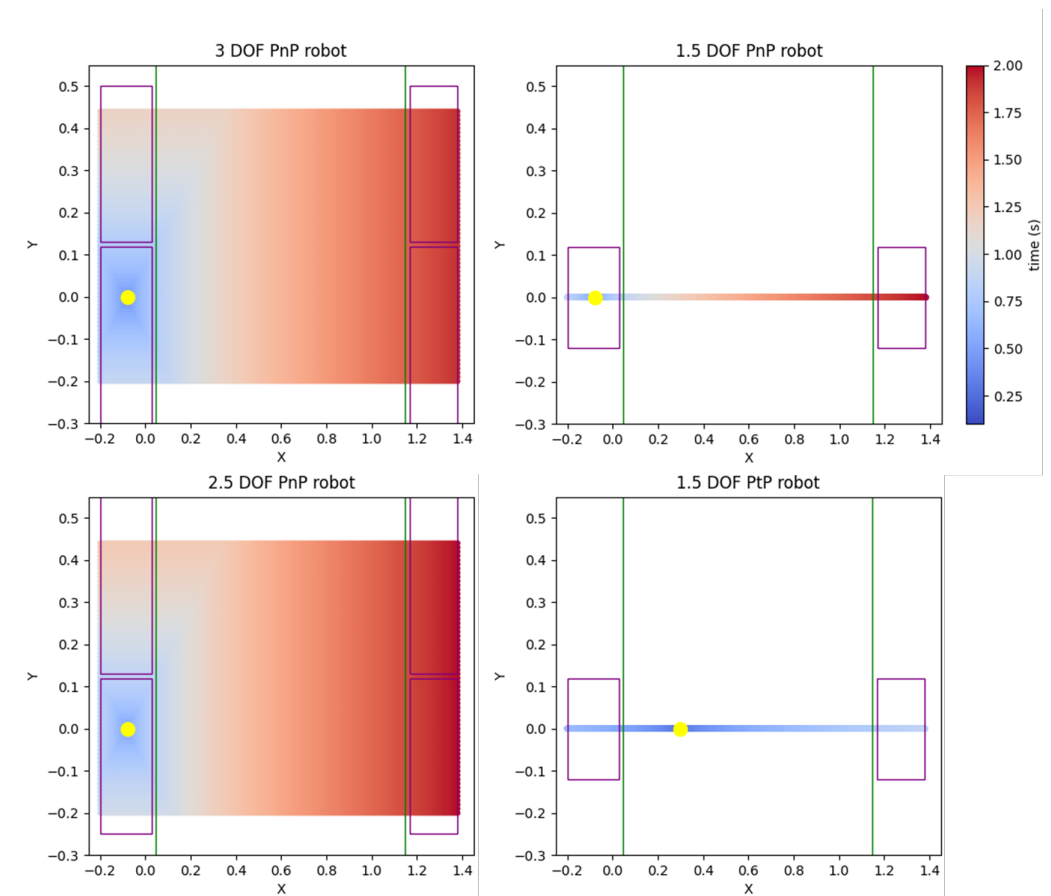
## 2.2. Process Time Estimation

The processing time (PT) is defined as the duration required for a robot to move from its initial position, grasp an object with the gripper, and remove it from the conveyor. The initial position is the last position from the previous pick. This location alters based on the following two considered scenarios:

1. **Pick-and-place (PnP) motion:** In this scenario, the robot moves from an initial position to the pick position and then returns to a position above a specified sorting bin to drop the object into the sorting bin. The location above the bin in which the waste is released defines the initial position of the next pick. This entire sequence of movements, from initial position to pick position to bin position, is referred to as a PnP motion.
2. **Pick-to-pick (PtP) motion:** Alternatively, if the gripper system allows for the disposal of the object into the bin without the robot returning to the bin, the motion profile is called a PtP motion. In this case, after picking an object, the grasping location becomes the initial position for the next pick, eliminating the need for the robot to move to the bin between picks. Usage of the HAVCO gripper facilitates the possibility of using the PtP motion profile.

The processing time (PT) can be estimated using forward and inverse kinematics [19]. Previous studies have employed forward and inverse kinematics to precalculate PTs to

reduce computation times for scheduling heuristics, particularly in robots with a more complex kinematic profile [20,21]. The time required for grasping or astricting an object depends on both the gripper and the characteristics of the object being grasped. For this reason, the grasping time needs to be either empirically determined or estimated with an algorithm based on the specific object and gripper combination. However, since no grasp-time-estimation algorithm is available to the knowledge of the authors, the grasping time is empirically determined. The grasping time varies between the different gripper types but is constant for a specific gripper over the different waste materials. The PT varies with the robot type due to differences in kinematic behavior among robot joints, which is influenced by the mechanical construction of the joints, the weight of the gripper, the mass of the picked object, and the maximum speed and acceleration of each joint. In this study, only Cartesian robots are used due to their low cost and the possibility of constructing robots with a large workspace. The use of Cartesian robots simplifies the kinematics to linear calculations. As a result, precalculated PTs are not necessary for this study. Figure 4 illustrates the PTs for the four different robot types: 1.5, 2.5, and 3-DOF robots with a PnP motion profile and a 1.5-DOF robot with a PtP motion profile. For the PnP motion, PTs are shown for movements starting from the yellow dot to a point in the workspace and returning to the yellow dot. For the PtP motion, PTs are shown for movements where the robot starts from the previous pick position (the yellow dot) and moves to a new point in the workspace to pick another object.



**Figure 4.** An overview of PTs for the 3-DOF PnP robot (left top), 2.5-DOF PnP (left bottom), 1.5-DOF PnP (right top), and 1.5-DOF PtP (right bottom). The green lines indicate the border of the conveyor. The purple boxes indicate the sorting bins, and the yellow dot indicates the initial position on which the robot started. (The x and y directions are expressed in meters.)

### 2.3. Sequence Planning Heuristic

The proposed algorithm is a collaborative multirobot task assignment heuristic that systematically considers the status of all available picking robots in the robotic sorting system. The general structure of the algorithm is outlined in Algorithm 1, with a more detailed discussion provided in the following paragraph. The heuristic is invoked each time a robot requests a new picking task, signified by an empty task queue. Initially, a set is generated containing objects that are either about to pass through or are currently within the requesting robot's workspace. Objects in this set are evaluated and sorted based on their estimated PT. Additionally, this evaluation incorporates the probability of successfully grasping each object, taking into account the specific gripper system installed on each robot. The evaluation process for multiple robots begins with the requesting robot, where the difference ( $\Delta_1$ ) in processing time between the shortest and the second shortest PT is calculated. Additionally, the difference ( $\Delta_2$ ) is calculated between the shortest PT after picking the object with the current shortest PT and the best object with the best PT if the second-best object was picked. These deltas are summed to assess whether selecting the best or second-best object offers a more favorable outcome. If the overall delta is negative, the object with the shortest PT is removed from the set of potential candidates, and the evaluation process restarts. If the overall delta is positive, the evaluation process considers the perspectives of other robots. If another robot's delta for the same object is higher, the requesting robot defers, and the object is removed from the set. This iterative process continues until an object is assigned to the requesting robot, ensuring efficient task allocation while considering the collaborative capabilities of the multirobot sorting system. The performance of the proposed heuristic is compared with traditional decision rules for PnP systems, specifically the First-In-First-Out (FIFO) and Shortest Processing Time (SPT) algorithms. The FIFO and SPT algorithms are widely adopted standard strategies in scheduling theory, where FIFO represents the simplest approach with minimal computational requirements and SPT is commonly used to minimize waiting time [22].

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#### Algorithm 1: Proposed heuristic.

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Input:  $O$ : a set of  $|O|$  objects on the conveyor belt
Output:  $O_{\text{best}}$ : the selected best object
 $O_{\text{best}} \leftarrow \emptyset$ ;
 $\text{EvaluateProcessingTime}(O)$  where  $O_i$  is the  $i^{\text{th}}$  ranked object based on processing time;
while true do
     $O_{\text{best}} \leftarrow O_1$ ;
     $\Delta_1 \leftarrow PT_2 - PT_1$  where  $PT_i$  represents the processing time of object  $i$ ;
     $\Delta_2 \leftarrow PT_1'' - PT_1'$  where  $PT_1'$  and  $PT_1''$  are the processing times if  $O_1$  or  $O_2$  are picked, respectively;
     $\Delta \leftarrow \Delta_1 + \Delta_2$ ;
    if  $\Delta < 0$  then
        Remove  $O_1$  from set  $O$ ;
         $O_{\text{best}} \leftarrow \emptyset$ ;
    end
    else
         $r \in R \text{ EvaluateProcessingTime}(O)$  where  $O_i^r$  is the  $i^{\text{th}}$  ranked object for robot  $r$ ;
         $\Delta_1^r \leftarrow PT_2 - PT_1$ ;
         $\Delta_2^r \leftarrow PT_1'' - PT_1'$ ;
         $\Delta^r \leftarrow \Delta_1^r + \Delta_2^r$ ;
        if  $O_{\text{best}} = O_1^r$  and  $\Delta < \Delta^r$  then
            Remove  $O_{\text{best}}$  from set  $O$ ;
             $O_{\text{best}} \leftarrow \emptyset$ ;
            break;
        end
    end
    if  $O_{\text{best}} \neq \emptyset$  then
        break;
    end
end
return  $O_{\text{best}}$ 

```

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### 3. Experiments

This study comprises three experiment sets designed to evaluate various aspects of the proposed heuristic and its performance in different scenarios. The first set of experiments aimed to determine the average efficiency of different grippers by testing them in a real waste sorting environment with a genuine waste stream. During the grasping experiment in the physical pr-MRF, the waste materials were fed, one by one, into the pr-MRF with a constant conveyor speed of 0.15 m/s. A simple grasp heuristic was used to position the gripper directly over the centroid of each object in a vertical orientation to perform the grasp. A grasp is considered successful when the waste object is successfully moved to the sorting bin. However, performing these experiments in the real pr-MRF introduce logistic complexity, hence the limited number of test results. A total of 272 picks were performed with the HAVCO gripper, 276 with the suction cup gripper, and 64 picks with the parallel gripper. The experiments with the parallel gripper are limited due to both the limited available time for experimental testing and the drastically lower performance compared to the alternative grippers. The resulting picking success rates are presented in Table 2. To estimate the grasp success of each gripper on specific waste materials in the subsequent simulation study, the average success rates obtained from this experiment for each of the gripper types is used.

**Table 2.** Gripper performance across different materials.

|               | Bottles | Cans | Foils | Tetrapack | Tray |
|---------------|---------|------|-------|-----------|------|
| HAVCO         | 30%     | 76%  | 100%  | 43%       | 48%  |
| soft_parallel | 66%     | /    | /     | /         | /    |
| suction_cup   | 77%     | 75%  | 100%  | 87%       | 64%  |

The second and third sets of experiments were performed in a simulation as described in Section 2.3. All of the tests were performed five times on the same instance due to the gripping uncertainty in the environment. The second set of experiments evaluated the performance of commonly used decision rules and the proposed heuristic on arbitrary test instances with varying bin configurations. The objective was to assess how system constraints, specifically those related to what a specific robot is allowed to pick, impact the effectiveness of task assignment algorithms. These constraints arise from restricting each robot's sorting capabilities to specific material classes. Due to limited space in the installation, the install of sorting bins for all different material classes was inhibited, limiting the available picking opportunities for each robot.

Since waste stream compositions are constantly changing and sorting systems are custom-built to specific use-case-related requirements, it is irrelevant to brute-force calculate the most efficient configuration. Instead, three distinct scenarios were analyzed with a different level of system constraint: In Scenarios 1 and 2, each robot had access to only two sorting bins, with each bin designated for specific waste fractions. The key difference between these scenarios lay in the allocation of materials to the bins and therefore the gripper types. In Scenario 1, the initial allocation of materials to bins was tested (Figure 1). Based on the obtained simulation results, the material allocations were reassigned in Scenario 2 to evaluate how different bin assignments affect system performance. In Scenario 2, the second robot grasped PET bottles and trays instead of foils and trays as in Scenario 1. In the third scenario, some of these imposed constraints were reduced by theoretically allowing each robot to sort a wider variety of materials, even when, in reality, the available space is too limited to realize this system. In this scenario, the HAVCO gripper focused on all the materials except for bottles, and the first two robots, equipped with a suction cup, also picked all the materials except for foils. This increased the potential for optimization and provided insights into the heuristic's performance under more flexible conditions.

Furthermore, in the last set of experiments, the conveyor belt velocity parameter within the simulation from the second scenario was varied to study its impact on the overall throughput of the system. The velocities ranged from 0.1 m/s to 0.3 m/s in increments of 0.05 m/s.

#### 4. Results

Table 2 presents the success rates of various grippers tested on different waste material classes.

The suction cup gripper demonstrated the best overall performance, followed by the HAVCO gripper. Due to the poor performance of the parallel gripper on objects it was thought to perform best on, possibly due to a misalignment in the grasp pose, it was not considered for further testing on objects other than bottles, as indicated by the “/” in Table 2. Consequently, the parallel gripper was replaced by the suction cup gripper in the computational experiments given the suction cup gripper’s superior performance. Despite the HAVCO gripper’s lower performance for certain object types, it remains relevant due to the reduced overall processing time by eliminating robotic motion (PtP motion), thus speeding up the waste sorting process.

As described in Section 3, the second set of experiments tested three distinct layouts with different task assignment algorithms. The performance metrics, detailed in Table 3, include the average success rate of picks and the picks per minute achieved by the systems during these tests.

**Table 3.** Average success ratio (%) and picks per minute (PPM) across different hardware scenarios at a conveyor belt speed of 0.15 m/s. The improvement (imp.) (%) of all scenarios over the FIFO approach of Scenario 1 is indicated as well.

|                | FIFO  | Imp.  | SPT    | Imp.  | Heuristic | Imp.  |
|----------------|-------|-------|--------|-------|-----------|-------|
| <b>Scene 1</b> |       |       |        |       |           |       |
| Success        | 84.0% | /     | 83.43% | −0.7% | 86.83%    | 3.4%  |
| PPM            | 192.7 | /     | 220.37 | 14.3% | 220.14    | 14.2% |
| <b>Scene 2</b> |       |       |        |       |           |       |
| Success        | 82.6% | −1.7% | 82.10% | −2.3% | 83.40%    | −0.7% |
| PPM            | 199.6 | 3.6%  | 224.19 | 16.3% | 225.48    | 17.0% |
| <b>Scene 3</b> |       |       |        |       |           |       |
| Success        | 78.8% | −6.2% | 79.23% | −5.7% | 87.04%    | 3.6%  |
| PPM            | 195.7 | 1.5%  | 230.07 | 19.4% | 242.62    | 25.9% |

The results indicate that both SPT and the proposed heuristic enhance overall system performance compared to the FIFO baseline. Specifically, the proposed heuristic achieves an improvement in picks per minute ranging from 14.2% to 25.9% with respect to the FIFO baseline depending on the constraints applied to the system by the bin configurations. Additionally, the average success rate of all the performed picks by the picking system increases up to 3.6% across different scenarios over the FIFO approach. By constraining the system through limiting the sorting capabilities of the robots with predefined materials for each bin, the limitations of the SPT rule are somewhat mitigated. However, when these constraints are lifted, like in Scenario 3, allowing the robots more flexibility to independently optimize their pick sequencing, the effectiveness of the SPT strategy diminishes. This decline is due to the SPT rule’s focus on maximizing the number of picks without considering pick success probability, leading to reduced overall efficiency as the SPT algorithm prioritizes quantity over quality in pick attempts. Furthermore, removing these system constraints also results in significantly higher PPM rates, as the robot can more effectively optimize their pick sequences, leading to faster and more efficient sorting operations.

The final experiment examines the impact of the conveyor belt velocity on the efficiency of the system. Since the same instance is used with different system parameters to allow the comparing of different approaches, the conveyor speed directly impacts the total mass of objects that is fed through the system in a fixed amount of time. Therefore, the total sum of the mass of all objects in the instance and the average of the system's makespan from the makespan of the tests with a specific planning approach and conveyor velocity are used to define the throughput in tons per hour. The results of this experiment are detailed Table 4.

**Table 4.** Average success rate (%), picks per minute (PPM), and their improvement (imp.) across different conveyor belt speeds or system throughput.

|                                | FIFO  | Imp.  | SPT   | Imp.  | Heuristic | Imp.  |
|--------------------------------|-------|-------|-------|-------|-----------|-------|
| <b>0.1 m/s<br/>0.36 ton/h</b>  |       |       |       |       |           |       |
| Success                        | 81.8% | /     | 81.4% | −0.5% | 82.4%     | 0.7%  |
| PPM                            | 185.0 | /     | 200.9 | 8.6%  | 201.7     | 9.0%  |
| <b>0.15 m/s<br/>0.54 ton/h</b> |       |       |       |       |           |       |
| Success                        | 82.6% | 1.0%  | 82.1% | 0.4%  | 83.4%     | 2.0%  |
| PPM                            | 199.6 | 7.9%  | 224.2 | 21.2% | 225.5     | 21.9% |
| <b>0.2 m/s<br/>0.72 ton/h</b>  |       |       |       |       |           |       |
| Success                        | 82.8% | 1.2%  | 81.9% | 0.2%  | 83.3%     | 1.9%  |
| PPM                            | 206.0 | 11.3% | 237.0 | 28.1% | 238.0     | 28.7% |
| <b>0.25 m/s<br/>0.90 ton/h</b> |       |       |       |       |           |       |
| Success                        | 82.6% | 0.9%  | 82.2% | 0.5%  | 84.1%     | 2.9%  |
| PPM                            | 210.2 | 13.6% | 247.6 | 33.8% | 248.5     | 34.3% |
| <b>0.3 m/s<br/>1.08 ton/h</b>  |       |       |       |       |           |       |
| Success                        | 83.4% | 1.9%  | 82.0% | 0.3%  | 84.2%     | 2.9%  |
| PPM                            | 216.4 | 16.9% | 251.8 | 36.1% | 254.4     | 37.5% |

The results indicate that higher conveyor velocities generally enhance system performance. Specifically, the proposed heuristic improves the average success rate of all the performed picks by 0.7% to 2.9% and increases PPM by 9% to 37.5% compared to the FIFO planning with a conveyor belt speed of 0.1 m/s. This increase in system throughput is directly attributed to the reduced downtime of the robots, leading to higher utilization rates. Initially, the increase in PPM is steep, but the rate of improvement gradually diminishes as conveyor belt velocity continues to rise, as is observable in Table 4. The PPMs do not increase linearly with the success rate; this is due to the fact that it is not only the success rate that has an impact on the PPM reached. Alternatively, the higher conveyor speed provides more opportunities to the scheduling algorithm to find a more efficient picking schedule. However, in this study, the gripper efficiencies are assumed to remain constant regardless of conveyor belt speed, although grasp efficiencies are anticipated to decline as the conveyor belt velocity increases. This reduction is caused due to the fact that the gripper can only be positioned close to the PPW product for a very limited time, limiting the force that can be generated by the gripper (in the case that the gripper needs time to build the grasping force, e.g., when vacuum or pneumatic pressure is used). The HAVCO gripper, in contrast, is less dependent on precise gripper positioning, allowing it to maintain prolonged contact in close proximity to the PPW object. This extended interaction time suggests that the HAVCO gripper is potentially less affected by increased conveyor velocities.

Overall, the findings in the system with the described assumptions highlight that the FIFO algorithm is relatively basic compared to the improvements achievable with the SPT and the proposed heuristic. In particular, the proposed heuristic demonstrates substantial

potential for enhancing robotic sorting and pick-and-place systems. However, the impact of the elevated conveyor speeds on the grasp efficiency of the different grippers still has to be determined. Aside from this, the grasp performance on more different waste materials, with different shapes, sizes and pollution levels, has to be determined.

## 5. Conclusions and Future Work

This work describes the development and evaluation of a novel heuristic for the scheduling of picks in a non-conventional robotic sorting system equipped with multiple different gripper systems and low-cost robots. Three different gripper systems are evaluated in a real-life pr-MRF on Crete. The HAVCO and finger gripper with passive-contact-driven fingers are unconventional grippers which were tested alongside the industry standard suction cup grippers to evaluate their performance on the PPW. A novel heuristic on the scheduling to improve efficiency of robotic sorting systems is investigated. The proposed heuristic specifically focuses on the average success rate of grasping operations and the pick rate in a PnP application for multirobotic waste sorting.

Three sets of experiments were conducted to assess the performance of different grippers, task assignment algorithms, and system configurations. The first experiment aimed to determine the grasp efficiencies of the three different grippers for the different PPW materials. The results revealed that the suction cup exhibited superior overall performance, followed by the HAVCO gripper. Due to the suboptimal performance of the parallel gripper, it was excluded from further testing. The results from this experiment informed the optimization of the proposed task assignment heuristic by providing information on grasp success rates. In the second set of experiments, the performance of the proposed heuristic was compared to those of two traditional decision rules (FIFO and SPT) under various sorting bin configurations. The finding demonstrated that increasing system flexibility, for example, by allowing robots to sort into a larger number of bins, significantly enhanced both the grasp success rate and the pick rate. The heuristic showed notable improvements in performance, with a pick rate increase of up to 25% and a rise in grasp success rate of up to 3.6% compared to the FIFO approach depending on the scenario. In the third set of experiments, the impact of conveyor velocity on system performance was analyzed. The results indicated that higher conveyor velocity generally improved pick rates due to higher robot utilization rates, while the overall share of picked objects gradually decreased.

Future work will focus on several key areas to further enhance robotic sorting systems. First, additional experiments will be performed to expand the dataset of grasp success rates across various materials and grippers. Alternatively, a grasp success prediction model can be investigated. This model could use an input, for instance, an image of an object, to predict the grasp success rate from a specific gripper on the object in the provided image. Second, considering the relatively low grasping success rates for some gripper types, improving gripper technology to increase grasp success rates and designing grippers for specific material streams could lead to more efficient sorting systems. Third, exploring the integration of variable conveyor belt velocity could offer dynamic adjustments (speed optimization) based on system load, optimizing performance in real time. However, for this, the impact of the conveyor speed on the gripper performance needs to be evaluated. Fourth, the material type, value, weight, and level of pollution could be used to assess the monetary value and avoided impacts on the environment by (improper) recycling for different sorting scenarios and efficiencies. These characteristics of each waste object could be used as an additional optimization parameter to prioritize the recovery of the most economically or environmentally valuable objects, especially in overloaded sorting systems where not all the material can be treated. Finally, investigating the potential for faster and more accurate



robotic components and conveyor systems as well as grasp planning algorithms could further boost sorting speed and efficiency, offering substantial opportunities to improve the overall economic viability and environmental performances of robotic sorting systems.

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## Abbreviations

The following abbreviations are used in this manuscript:

|           |                                                                    |
|-----------|--------------------------------------------------------------------|
| MDPI      | Multidisciplinary Digital Publishing Institute                     |
| PPW       | Plastic and packaging waste                                        |
| PnP       | Pick and place                                                     |
| MRF       | Material recycling facility                                        |
| PPM       | Picks per minute                                                   |
| pr-MRF    | Portable material recycling facility                               |
| HAVCO     | High Airflow Vertical CONveying                                    |
| TD-PnP-TW | Time-dependent pick-and-place scheduling problem with time windows |
| DOF       | Degree Of freedom                                                  |
| PT        | Processing time                                                    |
| PtP       | Pick to pick                                                       |
| FIFO      | First In First Out                                                 |
| SPT       | Shortest processing time                                           |

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